

CS 1101 Unit 7

Programming Assignment

Assignment Response

1. Dictionary Inversion

In a program, a dictionary contains lists of students and their courses. The teacher is interested to have a dictionary that has the courses as key and the students enrolled in each course as values.

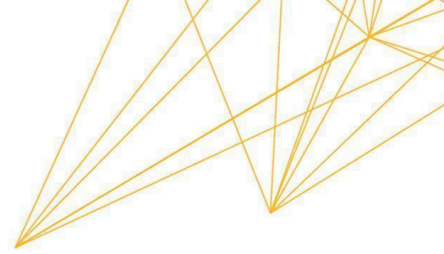
Each key has three different values.

To address this requirement, write a function to invert the dictionary and implement a solution that satisfies the teacher's need. In particular, the function will need to turn each of the list items into separate keys in the inverted dictionary. Also provide a technical explanation for the code and its output in minimum 200 words.

Sample input:

```
{  
  
    'Stud1: ['CS1101', 'CS2402', 'CS2001'],  
  
    'Stud2: ['CS2402', 'CS2001', 'CS1102']  
  
}
```

Inverted Output:

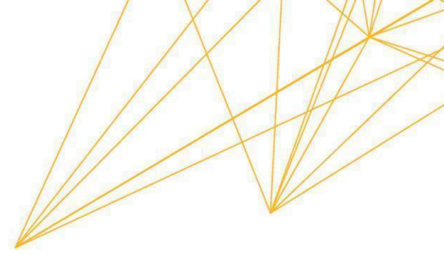


```
{  
  
    'CS1101': ['Stud1'],  
  
    'CS2402': ['Stud1', 'Stud2'],  
  
    'CS2001': ['Stud1', 'Stud2']  
  
    'CS 1102': ['Stud2']  
  
}
```

Programming Instructions:

1. Print the original dictionary as well as the inverted dictionary.
2. Include your the Python program and the output in your submission.
3. The code and its output must be explained technically. The explanation can be provided before or after the code, or in the form of comments within the code.

```
# Dictionary Inversion  
  
def invert_dictionary(original_dict):  
    """  
    Inverts a dictionary where values are lists.  
    The list items become keys, and the original keys become values.  
    """  
  
    invertedDict = dict()
```



```
# Iterate through the original dictionary using items()

for student, courses in original_dict.items():

    # Iterate through each course in the student's list

    for course in courses:

        # If the course is not yet a key, create a new list with the student

        if course not in invertedDict:

            invertedDict[course] = [student]

        # If the course already exists, append the student to the list

        else:

            invertedDict[course].append(student)

return invertedDict

# 1. Construct the original dictionary

studentsCourses = {

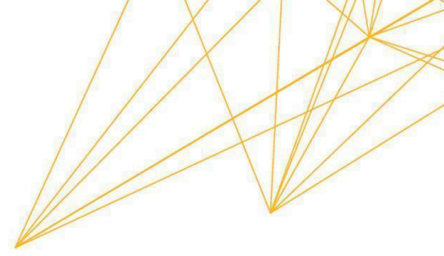
    'Stud1': ['CS1101', 'CS2402', 'CS2001'],

    'Stud2': ['CS2402', 'CS2001', 'CS1102']

}

# 2. Print the original dictionary

print("Original Dictionary:")
```



```
print(studentsCourses)

print("\n") # Adding a blank line for readability

# 3. Invert the dictionary and print the result

invertedResult = invert_dictionary(studentsCourses)

print("Inverted Output:")

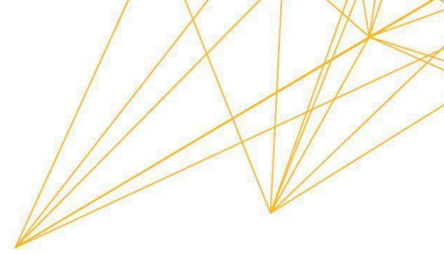
print(invertedResult)
```

```
• ~/Doc/UoPeopleCS1101-01ProgrammingFundamentals/Programming Assignment Unit 7 | main ?1 python3 dictionaryInversion.py
Original Dictionary:
{'Stud1': ['CS1101', 'CS2402', 'CS2001'], 'Stud2': ['CS2402', 'CS2001', 'CS1102']}

Inverted Output:
{'CS1101': ['Stud1'], 'CS2402': ['Stud1', 'Stud2'], 'CS2001': ['Stud1', 'Stud2'], 'CS1102': ['Stud2']}
```

Technical Explanation

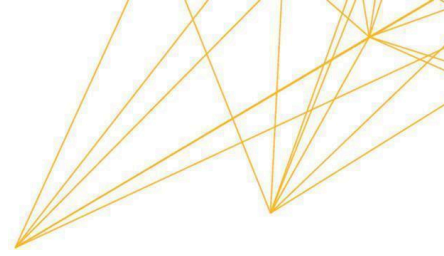
This Python program demonstrates how to manipulate and invert a complex data structure. As Downey (2015, p. 103) explains, *“a dictionary is like a list, but more general”* because in a list, indices must be integers, but in a dictionary they can be almost any type. According to W3Schools (n.d.), a dictionary is a *“collection which is ordered*, changeable, and does not allow duplicates.”* In the original dictionary, the keys are strings representing student names, and the values are lists representing their enrolled courses. Dictionary keys must be immutable types such as strings or tuples, but the values can be of any data type.



The `invert_dictionary()` function begins by creating an empty dictionary using `dict()`. To satisfy the requirement of making courses the keys and students the values, the program must traverse the entire data structure. As Raj (2022) explains, ***“When we reverse a given Python dictionary, we change the order of values in the key-value pair”***, so that the current key becomes the value and the current value becomes the key. Because our values are lists, this requires nested for loops. The outer loop uses the `.items()` method, which Raj (2022) notes can be used to ***“take each item from the input dictionary one by one”*** - returning each student and their course list as a pair. The inner loop then iterates through each individual course in that student's list.

During the inner traversal, the program uses the `in` operator to check whether a course already exists as a key in `invertedDict`. As Downey (2015, p. 104) notes, ***“the `in` operator takes about the same amount of time no matter how many items are in the dictionary”*** because Python dictionaries are implemented using a hashtable. If the course is not yet a key, the program creates a new entry initialized with a singleton. Downey (2015, p. 112) defines a singleton as ***“a list (or other sequence) with a single element”*** - in this case, a list containing just the current student. If the course already exists, `.append()` adds the new student to the existing list. This is possible because, as W3Schools (n.d.) states, ***“Dictionaries are changeable, meaning that we can change, add or remove items after the dictionary has been created.”***

Examining the printed output confirms the logic works correctly. The original dictionary shows two students each enrolled in three courses. In the inverted output, 'CS1101' maps only to ['Stud1'] because only Stud1 is enrolled in it. Both 'CS2402' and 'CS2001' map to ['Stud1',



'Stud2'] because both students share those courses - the `.append()` logic correctly added Stud2 without overwriting Stud1. Finally, 'CS1102' maps only to ['Stud2']. This output confirms the inversion is complete and accurate.

446 words

References

Downey, A. (2015). Think Python: How to think like a computer scientist. Green Tree Press.

<https://greenteapress.com/thinkpython2/thinkpython2.pdf>

Raj, A. (2022, March 3). Reverse a dictionary in Python. PythonForBeginners.

<https://www.pythonforbeginners.com/basics/reverse-a-dictionary-in-python>

Python Dictionaries. (n.d.). W3schools.

https://www.w3schools.com/python/python_dictionaries.asp